运动方程：

$$mn_0\left[\frac{\partial \vec{v}_e}{\partial t}+(\vec{v}_e\cdot\nabla)\vec{v}_e\right]=-e n_e\vec{E}-\nabla\vec{P}$$

$$Mn\left[\frac{\partial \vec{v}_i}{\partial t}+(\vec{v}_i\cdot\nabla)\vec{v}_i\right]=-e_n\nabla\phi-\gamma_i K_B T_i\nabla n$$

$\vec{u}=\vec{u}_0+\vec{u}_1$，

$$-i\omega Mn\frac{\partial \vec{u}_1}{\partial t}=-en_i k\phi_i-\gamma_i K T_i\, ik n_{i1}$$

连续性方程：

$$\frac{\partial n}{\partial t}+\nabla\cdot(n\vec{v})=0.$$

$$\frac{\partial n_i}{\partial t}+n_0\nabla\vec{v}_{i1}=0$$

$$i\omega n_i-n_{0i}kv_{i1}=0$$

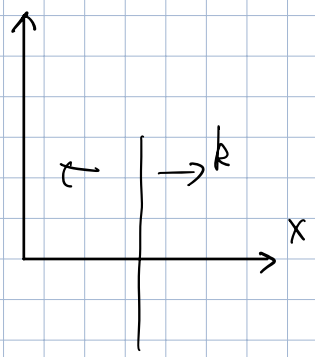
假设电子热平衡，电子满足玻尔兹曼分布：（将速度积分）

$$n=n_0\exp\left(\frac{e\phi}{kT_e}\right)$$

$$=n_0\left[1+\frac{e\phi_1}{kT_e}+\frac{1}{2!}\left(\frac{e\phi_1}{kT_e}\right)^2+\cdots\right]$$

准中性条件：$n_{i1}=n_{e1}$

$$\frac{\omega^2}{k^2} = \frac{KT_e + \gamma_i k T_i}{M} = \frac{d\omega}{dk}$$



$$-\nabla^2\phi = \frac{e(n_i - n_e)}{\varepsilon_0}$$

$$k^2\phi^2 = \frac{e}{\varepsilon_0}\left(n_{i1} - \frac{e\phi_1}{KT_e} n_0\right)$$

$$\phi_1\left(k^2 + \frac{en_0^2}{\varepsilon_0 k T_e}\right) = \frac{e}{\varepsilon_0} n_{i1} \qquad \lambda_D^2 = \frac{\varepsilon_0 k T_e}{n_0 e^2}$$

$$\phi_1\left(k^2 + \frac{1}{\lambda_D^2}\right) = \frac{e}{\varepsilon_0} n_{i1}$$

$$i\omega M n_0 v_{i1} = en_1 k \frac{en_{i1}}{\varepsilon_0\left(k^2 + \frac{1}{\lambda_D^2}\right)} + \gamma_i k T_i \, ik \frac{n_i k}{\omega}$$

$$\frac{\omega^2}{k^2} = \frac{1}{M}\left(\frac{KT_e}{(1 + k^2\lambda_D^2)} + \gamma_i K T_i\right)$$

$$\left(\frac{\lambda_D}{\Lambda}\right)^2 \qquad 1$$